ECOS-EEG: Evolutionary Channel Optimization for Subject-Independent EEG Emotion Recognition

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***Abstract*—Electroencephalography is a direct measure of neural activity and has been developed in large-scale application to the affective state analysis because of the high temporal sensitivity and resistance to behavioral masking. The differences in spectral and dynamical properties of cortical signals are linked to valence and arousal dimensions of emotion. However automatic recognition is difficult due to the inter-subject variability, high sensitivity to noise, and spatial redundancy with a high level of density between electrodes. The paper suggests the idea of Evolutionary Channel Optimization System(ECOS-EEG). A feature-based framework, which is achieved in an evolutionary channel optimization approach using a combination of differential entropy, band power, Hjorth parameters, statistical measures, and frontal asymmetry to feature pattern to determine an informative subset of electrodes. The proposed framework reduced the number of channels from 32 to 14.A 56% reduction in channels and reduces spatial redundancy but retains discriminative information. The feature representation produced is then classified in a radial basis function support vector machine after standardization in a pooled subject independent environment. The results of preprocessed DEAP data on 80/20 stratified split with 5 threshold as binary classifications of valence and arousal reveal accurate performance of 69.41% and 71.14% respectively, showing that structured spatial optimization with small feature decompositions can yield comparable performance with reduced number of features without the complexity of deep neural networks.**

***Index Terms*—EEG emotion recognition, subject-independent learning, channel optimization, differential entropy, support vector machine.**

I. Introduction

Emotion is among the key factors in human cognition, decision-making, as well as social interaction, affecting perception, memory, and behavior responses. Recognition of automatic emotions has thus been of considerable interest in affective computing, with facial expressions, speech signals and physiological measurements being the most commonly studied modalities. Electroencephalography (EEG) is one of them and is a direct measure of cortical activity which is highly temporally localized and opposes voluntary masking, and this makes them most applicable to study internal affective states [1], [2]. Valence and arousal dimensional scale of emotion has gained wide applicability to EEG-based modeling of emotion because it is interpretable and fits benchmarking datasets like DEAP [3].

EEG-led emotion recognition is technically difficult, regardless of its benefits. Neural signals are characterized by low signal-noise ratio, non-stationary and heavy inter-subject variability which makes it challenging to extract stable patterns of discrimination [4]. Moreover, the traditional multi-channel recordings add a spatial redundancy and correlated electrodes add redundant information that can compromise the generalization results. The use of feature-based informatics with differential entropy, spectral power, Hjorth parameters and statistical descriptors have been popular to extract informative characteristics of signals at the expense of eliminating variations in raw signals [5], [6]. Nevertheless, most of the current implementations are based on fixed full-channel settings that do not explicitly combat spatial redundancy.

Recent developments in deep learning have seen better performance with the use of hierarchical representations learned directly on EEG signals. Topographic feature maps and spatial filtering mechanisms based convolutional neural networks have been shown to have competitive results on DEAP when performing valence-arousal classification [7], [8]. Transformer and self-supervised models also strive to achieve subject-independent robustness by means of attention and auxiliary goals [9], [10]. Though capable of capturing quite complicated spatial-temporal interactions, these kinds of architectures tend to be heavily parameterized, computationally expensive, and vulnerable to small training surrogates, which exacerbates the problem of overfitting when cross-subject.

The use of subject-independent emotion recognition is also a source of further complexity since distributional changes by participants play a crucial role in influencing the stability of the classifier. Strategies of domain adaptation and adversarial learning have been proposed in order to minimize inter-subject discrepancies and also enhance transferability , [11]. Although these techniques improve generalization, they tend to prioritize the representation alignment as opposed to structural efficiency of the spatial dimension, and the problem of channel-level redundancy is inadequately investigated in the compact feature-driven pipelines.

Based on these empirical findings, the paper tackles the problem of spatial redundancy and subject-independent generalization on a classical feature-based framework. As an alternative to making architecture more complex, we explore ways of improving discriminative efficiency by optimizing the number of electrodes without compromising interpretability. The proposed ECOS-EEG model combines spectral and statistical measures with a mechanism of channel optimization

in evolutionary channel to determine the informative set of electrodes used to classify valence and arousal on the pools in an informative sub-set of electrodes and a pool-independent manner.

The rest of this paper includes methodological formulation of ECOS-EEG and experimental assessment applied on the processed DEAP data and comparative analysis with the representative methods. These findings represent that professional space optimization on a small scale learning system gives competitive results with computational simplicity.

The main contribution of this paper are as follows:

1) Development of ECOS-EEG, a compact feature-based framework for subject-independent valence–arousal classification.
2) Evolutionary channel optimization reduced 32 electrodes to an optimized 14-channel subset to address spatial redundancy and complexity.
3) Competitive performance on the DEAP dataset using an SVM-based architecture under pooled subject-independent evaluation.

The rest of the paper is arranged in the following manner. Section II is a review of related literature on EEG-based emotion recognition and subject independent modeling. Section III supports the proposed ECOS-EEG framework and the process involves feature extraction together with optimization of the channels in evolution. IV outlines the experiment design and test protocol. Section V includes the description of the analysis of performance and comparative outcomes. Lastly, Section VI summarizes the paper and gives future research directions.

II. Related Work

Preliminary studies in EEG-based affect analysis led to the conclusion that neural oscillatory processes encode affective state discriminative information, showing an association between spectral features and affective dimensions of valence and arousal [1], [2]. These fundamentals inspired widespread investigation into feature engineering approaches to identifying informative signal properties and reduction of noise and non-stationarity effects of EEG data records [4].

Classical methods were mainly based on handcrafted descriptors based on time-domain and frequency-domain analysis. They have been very common place because differential entropy, spectral power estimations, Hjorth parameters and higher-order statistical metrics have been shown to be able to compactly describe neural variability and band specific dynamics [5], [6]. They are usually associated with machine learning classifiers like support vector machines or linear discriminant analysis, and they are used to do valence-arousal discrimination. Although such pipelines are efficient in computation and also understandable, most implementations provide fixed electrode formations, and do not formally consider redundancy created by dense multi-channel recordings. Consequently, overlapping information between correlated channels might be involved, which can restrict generalization in the subject- independent [12].

The advent of deep learning has made a tremendous impact on the study of EEG emotion recognition. DEAP has been shown to perform better with convolutional neural networks which convert electrode signals to spatial topographic maps by learning local spatial dependencies across channels [7], [8]. The architectures facilitate vixous levels of feature learning and permitting end-to-end learning, dispensing on presupposed distinction-crafting. Other more sophisticated models include temporal modeling and attention processes to represent dynamic dependencies and focusing on salient spatial regions [9].For instance, Rajpoot et al. proposed an attention-driven deep learning framework combining convolutional and recurrent architectures for subject-independent EEG emotion recognition, demonstrating improved performance through learned spatial–temporal representations [10]. Moreover, there are self-supervised and auxiliary learning techniques that have been proposed to assist the robustness of representation and the use of people to improve transferability [10]. Although these developments have taken place, deep neural models can be very parameterized and hard to compute, which can result in overfitting on small or heterogeneous training data.While deep learning models achieve strong performance by learning complex spatial–temporal representations, they typically require full-channel inputs and involve high computational complexity. In contrast, the proposed ECOS-EEG framework focuses on a compact and interpretable pipeline by leveraging handcrafted features and optimized channel selection, making it more suitable for efficient and practical EEG-based emotion recognition.

Several studies have explored channel selection and electrode reduction techniques to address redundancy in multi-channel EEG recordings. Earlier works employed classical machine learning approaches to identify informative subsets of electrodes, demonstrating that a reduced number of channels can still provide meaningful emotion classification performance [13]. More recent studies have further shown that channel selection can significantly improve efficiency while maintaining recognition accuracy, highlighting the presence of redundant and less informative channels in dense EEG setups [14] . Additionally, advanced approaches have integrated optimization strategies with deep learning models, such as the ACO–CNN–LSTM framework, which dynamically selects optimal channels and achieves high accuracy with lightweight data representations [15]. These findings emphasize the importance of spatial optimization in EEG-based emotion recognition. however, many existing methods rely on complex architectures, while compact feature-based frameworks remain relatively underexplored.

A major challenge continues to be generalization over subjects since responders vary in their neural responses inter-individually. Domain adaptation and adversarial learning methods have been proposed to cut distributional distances between participants by matching latent features spaces [16]. The goal of these approaches is to improve the cross-subject performance through learning invariant representations, but they tend to be performed full-channel-wise and pay more attention to the architectural complexity of an inherent way instead

of structural reduction of the spatial dimension. Therefore, despite the prominent advancements that have been made in representation learning and transfer modeling, more systematic optimization of electrode subsets in compact feature-driven models has experienced relatively lower levels of attention.

III. Proposed ECOS-EEG Framework

The ECOS-EEG proposal aims at optimizing emotion recognition in subjects independent of the cognitive tasks by improving the spatial arrangement of EEG electrodes in a compact feature-based framework. The approach combines handcrafted spectral and statistical descriptors with an evolutionary channel optimization approach, as opposed to using full-channel representations, to find an informative subset of electrodes. Computation of features is performed across all channels, and candidate channel subsets are evaluated using a support vector machine classifier with a pooled subject-independent protocol. The general aim of the strategy is to minimize spatial overlap and retain discriminative information regarding valence and arousal. Fig 1 illustrate the framework of Evolutionary Channel Optimization for Subject Independent EEG Emotion Recognition. [h]

A. Feature Extraction

Features are calculated separately for each channel of an EEG trial. The extraction techniques include differential entropy (DE), power spectral density (PSD), Hjorth parameters, and higher-order statistical measures. The theta (4–8 Hz), alpha (8–14 Hz), beta (14–31 Hz), and gamma (31–50 Hz) frequency bands are derived using a fourth-order Butterworth band-pass filter, and variance-based entropy is computed for each band. The power spectral density is obtained using the Welch method with a fixed segment length, and band-wise power values are calculated accordingly. Hjorth parameters—activity, mobility, and complexity—are computed to characterize the temporal dynamics of the signal. Additionally, skewness and kurtosis are calculated to capture higher-order statistics.

Inter-hemispheric difference features are calculated for symmetric channel pairs whenever both electrodes are selected, providing hemispheric asymmetry information. Such asymmetry representations capture lateralized neural responses related to affective processing. All feature computations are performed prior to channel optimization, enabling efficient evaluation of candidate subsets via channel indexing.

Mathematical Formulation

For a band-limited EEG signal with variance σ^2, differential entropy(DE) is computed as

$$DE = \frac{1}{2}\log\left(2\pi e\sigma^2\right). \tag{1}$$

Here P_B denote the band power within a frequency interval B, computed from the PSD obtained via Welch's method.

$$P_B = \frac{1}{|B|}\sum_{f\in B} P(f), \tag{2}$$

Here $|B|$ represents the number of frequency bins within band B.

Hjorth parameters are calculated as

$$\text{Activity} = \text{Var}(x), \tag{3}$$

$$\text{Mobility} = \sqrt{\frac{\text{Var}(\dot{x})}{\text{Var}(x)}}, \tag{4}$$

$$\text{Complexity} = \frac{\text{Mobility of } \dot{x}}{\text{Mobility of } x}$$

here $\dot{x}$ and $\ddot{x}$ denote first- and second-order temporal differences of the signal, respectively.

Higher-order statistical descriptors are computed as

$$\text{Skewness} = \frac{\mathbb{E}[(x-\mu)^3]}{\sigma^3}, \tag{5}$$

$$\text{Kurtosis} = \frac{\mathbb{E}[(x-\mu)^4]}{\sigma^4}. \tag{6}$$

For symmetric electrode pairs, inter-hemispheric asymmetry is defined as

$$FAA = F_L - F_R, \tag{7}$$

Here FAA denotes frontal alpha asymmetry, F_L and F_R denote feature values extracted from corresponding left and right hemisphere channels.

In the above formulations, x represents the discrete EEG signal of a single channel, $\text{Var}(\cdot)$ denotes variance, μ is the mean of the signal, σ is the standard deviation, and $\mathbb{E}[\cdot]$ denotes the expectation operator.

B. Evolutionary Channel Optimization

An optimization problem of choosing a channel is modeled as a binary optimization problem where every possible solution is modeled as a mask vector.

$$Mi \in \{0,1\}^{32}, \tag{8}$$

Here Mi[j] = 1 denotes that electrode number j is chosen and 0 otherwise.The number of individuals $P = 11$ and iterate over $G = 50$generations. The aim is to come up with an optimal channel subset C* which maximizes the accuracy of the classification.

Here F represents the already-computed feature Tensor based on the EEG data X = 0.1: N x 32 x T. In each case of individual M_i, the selected-channel features are calculated by indexing F on the active entries of M_i. The characteristics of inter-hemispheric asymmetry are added in conditions of symmetrical pairs of electrodes. This feature proportionate is then assessed with an 80/20 stratified split and a support machine with radial basis are trained. The measure of the accuracy of the test A_i is the fitness value of that individual.

Each generation has a process of identifying the individual with the best performance as.

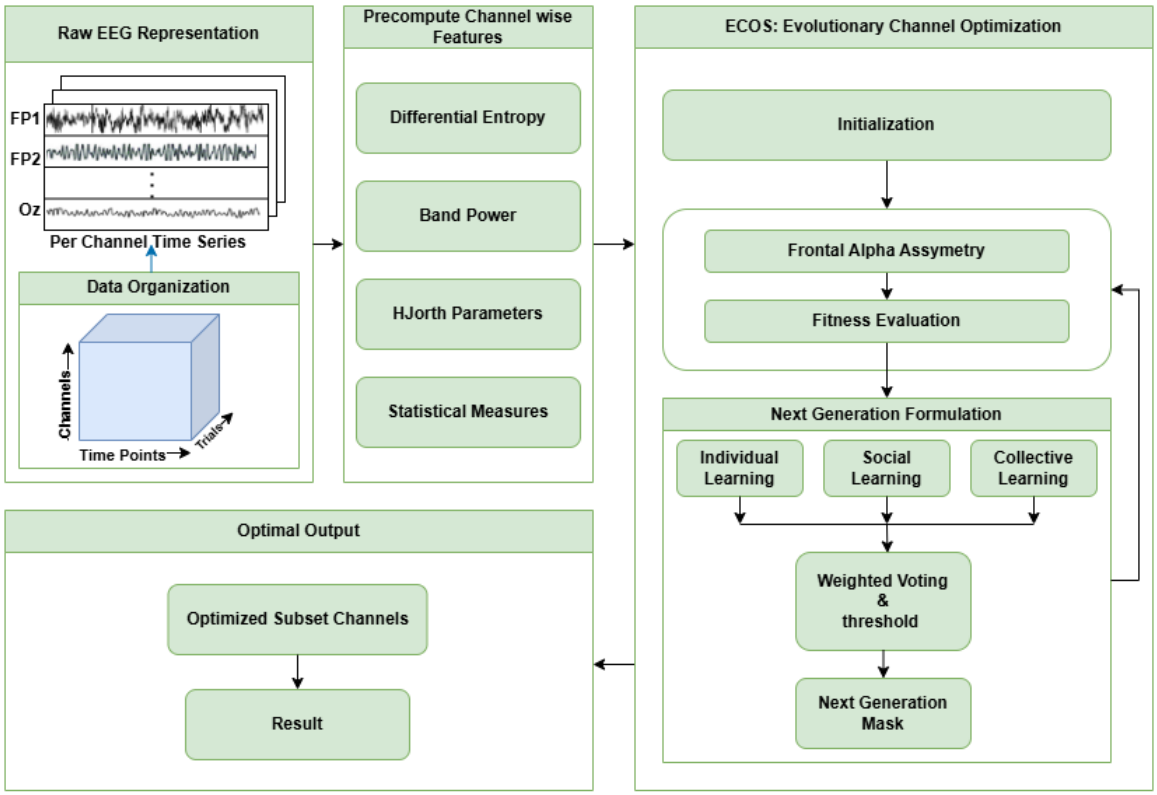


Fig. 1. ECOS EEG: Evolutionary Channel Optimization for Subject Independent EEG Emotion Recognition Framework

$$i^\star = \arg\max_i A_i. \tag{9}$$

When $A_{i^\star}$ is better than the current best accuracy of $A^\star$, then the global best mask and fitness are updated as

$$C^\star \leftarrow M_{i^\star}, \qquad A^\star \leftarrow A_{i^\star}. \tag{10}$$

In order to control the process of population development, a Social Learning mask, denoted as SL-in in the set of 0 and 1 in 32 bit, is resolved at each generation by majority vote by all individuals. For each channel index j,

$$SL[j] = \begin{cases} 1, & \text{if the majority of individuals select channel } j, \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

Three learning components are used in forming the next generation:

1) Individual Learning (IL): A candidate mask is obtained by using mutation on a randomly chosen individual, to yield exploratory random variations in the channel configuration.
2) Social Learning: Selecting channels according to the whole group behavior is the feature of Social Learning (SL).
3) Collective Learning (CL): There is a global best mask, which is denoted by $C^\star$ that provides an indication on the way to historically optimal configurations.

For a channel index j, let IL_j, SL_j, and $C_j^\star$ denote the Individual Learning, Social Learning, and global best (Collective Learning) values, respectively. The weighted vote is computed as

$$V_j = w_{IL} \cdot IL_j + w_{SL} \cdot SL_j + w_{CL} \cdot C_j^\star, \tag{12}$$

where

$$w_{IL} + w_{SL} + w_{CL} = 1. \tag{13}$$

The new mask entry is determined using a threshold rule:

$$M_j^{new} = \begin{cases} 1, & \text{if } V_j \geq \tau, \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

Each channel position is assigned a weighted voting process which sums IL, SL and $C^\star$, and a threshold operation is used to identify channels which are included in the new mask. The two highest ranked are maintained (elitism). Further diversity control as well as injection mechanisms is implemented to ensure that convergence is not experienced too soon and explore search space.

Once there have been G generations the algorithm returns the optimized channel subset according to which the highest classification accuracy has been observed as $C^\star$.

Algorithm 1 ECOS-EEG: Evolutionary Channel Optimization

Input: EEG tensor $X \in \mathbb{R}^{N\times 32\times T}$, labels y
Input: Number of individuals $P = 11$, iterations $G = 50$
Output: Optimal channel subset $C^\star$

$F \leftarrow \textsc{PrecomputeFeatures}(X)$
Initialize population $M = \{M_i\}_{i=1}^{P}$, where $M_i \in \{0,1\}^{32}$
Initialize $A^\star$
for $g = 1$ to G **do**
  for each $M_i \in M$ **do**
    $X_i \leftarrow \textsc{SelectChannels}(F, M_i)$
    $X_i \leftarrow \textsc{AddAsymmetry}(X_i, M_i)$
    Split (X_i, y) into train/test (80/20)
    Train SVM (RBF, $C = 10$, $\gamma =$ scale)
    $A_i \leftarrow$ test accuracy
  end for
  $i^\star \leftarrow \arg\max_i A_i$
  if $A_{i^\star} > A^\star$ **then**
    $C^\star \leftarrow M_{i^\star}$
    $A^\star \leftarrow A_{i^\star}$
  end if
  Compute social learning mask via majority voting
  Preserve top two individuals
  for remaining individuals **do**
    Generate individual learning mask via mutation
    Combine masks and apply threshold
  end for
  Apply diversity control and injection
end for
return $C^\star$

C. Classification Strategy

A support vector machine (SVM) with a radial basis function (RBF) kernel is employed for classification. Feature vectors are normalized using the training split statistics, and the same normalization is applied to test data. Model parameters are fixed at $C = 10$ and $\gamma =$ scale. Valence and arousal are treated independently, each as a binary classification task with a threshold of 5 converting dimensional ratings into binary labels. Performance is reported under a pooled subject-independent evaluation protocol.

Algorithm 1 presents the workflow of the proposed method, detailing the process from EEG feature extraction to optimized channel selection and classification. It describes how informative channels are iteratively identified and evaluated to improve performance. The final output consists of the selected channel subset and classification results.

IV. Experimental Evaluation

The preprocessed DEAP dataset is used in experiments having EEG recordings of 32 individuals exposed to 40 one-minute stimuli in the form of music videos.EEG signals are sampled at 128 Hz. There are emotional annotations on a 9-point scale on valence, arousal, dominance and liking.The dataset consists of five EEG frequency bands: delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–45 Hz). The threshold on which binary classification is carried out is set at 5, according to which ratings above 5 are considered as high and the rest as low. Valence is modelled independently as is arousal.

The precomputed feature tensor is indexed to get features of the selected channel subsets. In cases where the electrodes are symmetrical in pairs, asymmetry characteristics are introduced between the hemispheres. Stratified sampling with random seed is used to divide the data into the 80% training and 20% testing sets. Standardization of features is done based on statistics obtained on the training set and then implemented on the test data.

TABLE I
Comparison on the DEAP Dataset

Model	Channels	Valence	Arousal
CNN-based Model	32	66.73	68.28
Deep Neural Network	32	71.09	72.58
Adversarial Domain Adaptation	32	76.48	69.72
TOPO-FM CNN	32	76.30	76.54
ECOS-EEG (Proposed)	**14**	**69.41**	**71.14**

Table I compares the proposed ECOS-EEG framework to the current approaches to the DEAP dataset. Although certain deep learning models are more accurate, the suggested methodology competitive results are obtained using a significantly simpler pipeline. In contrast to the state-of-the-art approaches, ECOS-EEG is not based on deep learning to extract features or classify them, but relies on handcrafted features and SVM as a classifier. It also cuts down the channels to 14 instead of 32 by 56 percent, yet with acceptable performance. This demonstrates a trade-off between accuracy and practicality of models in terms of simplicity, efficiency and interpretability.Despite lower accuracy than deep models, the proposed method significantly reduces computational complexity and electrode dependency, making it suitable for real-world and wearable EEG systems.

The classification is performed with the help of a kernel support vector machine of radial basis functions with the c = 10 and gamma = scale settings. The test accuracy is used as evaluation metric. The channel subsets with the optimizations of both tasks have 14 electrodes. The selected channels for valence are F3, F7, CP5, CP1, F2, P7, Oz, Pz, AF4, Fz, F4, FC2, CP2, and P8 with an accuracy of 69.41% and for arousal are Fp1, F3, FC1, T7, CP1, P7, PO3, Pz, Fp2, AF4, F4, FC6, P4, and PO4 are selected with an accuracy of 71.14% with a 56% reduction in channels for both cases.

The evaluation also compared with alternative classifiers but the support vector machine showed the best generalization behaviour and hence it was chosen as the final model.

TABLE II
ABLATION STUDY: CLASSIFIER COMPARISON ON OPTIMIZED CHANNEL SUBSETS

Classifier	Valence Accuracy	Arousal Accuracy
LightGBM	58.20	62.89
XGBoost	58.20	64.06
LDA	66.41	62.11
SVM (RBF)	**69.41**	**71.14**

Table II shows the results of various classifiers tested on the optimized subsets of channels. Experiments are all performed under identical preprocessing, feature extraction and train-test conditions to facilitate a fair comparison. It was found that the SVM with an RBF kernel is the best among other classifiers like LightGBM, XGBoost, and LDA in terms of generalization potential, which demonstrates its better ability to generalize in the context of the given feature representation and means that it is the best choice in the proposed framework.

V. LIMITATIONS AND FUTURE WORK

The proposed ECOS-EEG framework demonstrates competitive performance with reduced channel complexity; however, several limitations remain. First, the model does not achieve the highest accuracy compared to recent deep learning approaches, which are capable of learning more complex spatial–temporal representations from EEG signals. Second, the evaluation is limited to a specific dataset and experimental setup, which may restrict the generalizability of the results across different recording conditions. Additionally, the current framework relies on handcrafted feature extraction, which may not fully capture all underlying neural dynamics.

In future work, the framework will be extended to multiple datasets to improve generalization and robustness. Furthermore, hybrid approaches that combine channel optimization with deep learning-based feature representation will be explored to enhance performance while maintaining efficiency. Finally, real-time implementation and validation on wearable EEG systems will be investigated to assess the practical applicability of the proposed method.

CONCLUSION

This research introduced ECOS-EEG, a feature-oriented system to classify EEG-based valence and arousal based on the spatial optimization principle as opposed to the architectural sophistication. The proposed method applied by taking into consideration variations of the entropy, spectral characteristics and Hjorth parameters, statistical measures and asymmetry among the hemispheres selects a 14-channel subset of the original 32-electrode system through an evolutionary channel selection strategy by incorporating concepts of differential entropy, spectral characteristics and Hjorth parameters, statistical measures and inter-hemispheric asymmetry. A radial basis function support vector machine is used to test the resulting compact representation of subject independent 80/20 split on the preprocessed DEAP data.

Experimentally, it is shown that the choice of an orderly electrode enhances generalization with only computational simplicity, with 69.41% valence and 71.14% arousal classification while reducing the number of channels by 56%. The findings suggest that even as a feature pipeline, spatial reduction can be performed competitively without using deep neural structures. Further optimization strategy development and cross-dataset validation are the areas of work that will be carried out in the future to better understand the optimization strategy and its robustness.

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